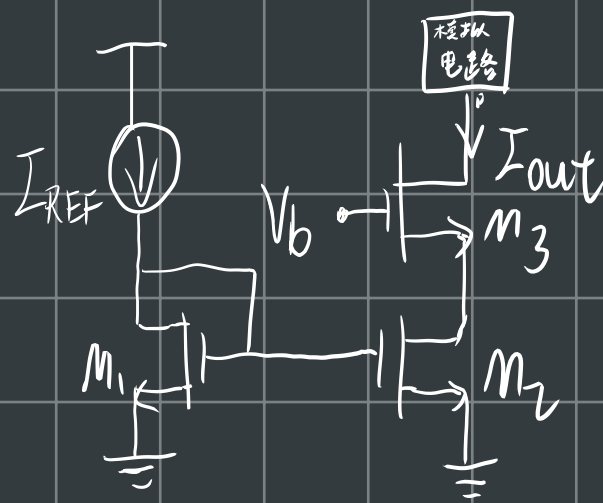


5.5 假设 $(W/L)_{1,3} = 40/0.5$, $I_{REF} = 0.3 \text{ mA}$ 且 $r = 0$

(a) 确定使 $V_x = V_y$ 时的 V_b

(b) 如果 V_b 偏离(a)中计算的值 100 mV , I_{out} 与 I_{REF} 之间的不匹配是多少?

(c) 如果电路用共源共栅电流源作负载, 使 V_b 变化 1 V 时, V_y 将变化多少?



(a) $V_b = 2V_{GS1}$ (反证法)

$$V_{GS1} = V_{th1} + \sqrt{\frac{2I_{D1}}{K'_1 \left(\frac{W}{L}\right)_1}}$$

$$= 0.7 \text{ V} + \sqrt{\frac{2 \times 0.3 \text{ mA}}{\frac{110 \mu\text{A}}{\text{V}^2} \times \frac{40}{0.5}}} = 0.96 \text{ V}$$

$$V_b = 1.92 \text{ V}$$

$$(b) \begin{cases} I_{REF} = \frac{1}{2} K'_1 \left(\frac{W}{L}\right)_1 (V_{GS1} - V_{th1})^2 (1 + \lambda_1 V_{GS1}) \\ I_{out} = \frac{1}{2} K'_2 \left(\frac{W}{L}\right)_2 (V_{GS2} - V_{th2}) (1 + \lambda_2 V_{GS2}) \end{cases}$$

$$\Rightarrow I_{out} = \frac{1 + \lambda V_{GS2}}{1 + \lambda V_{GS1}} I_{REF}, \quad (\lambda_1 = \lambda_2)$$

$$= \frac{1 + \lambda (V_{GS1} - \Delta V_b)}{1 + \lambda V_{GS1}} I_{REF}$$

$$= \frac{1 + 0.04/\text{V} (0.96 \text{ V} - 0.1 \text{ V})}{1 + 0.04/\text{V} \times 0.96 \text{ V}} I_{REF}$$

$$= 0.996 I_{REF}$$

$$\therefore \frac{|I_{out} - I_{REF}|}{I_{REF}} \times 100\% = 4\%$$

(C)

$$\Delta V_T = \frac{\Delta V_P}{(g_{m3} + g_{mb3}) r_{o3}}$$

$$= \frac{1V}{\sqrt{2 \times \frac{110\mu A}{V^2} \times \frac{40}{0.5} \times 3mA \times \frac{0.04}{V} \times 0.3mA}}$$

$$= 5.2mV$$

$$\therefore V_T = V_{GS1} + \Delta V_T = 0.96V + 0.0052V \\ = 0.9652V$$